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NATURAL CONVECTION IN DIRECTIONAL DENDRITIC SOLIDIFICATION OF METALLIC ALLOYS—I. MACROSCOPIC EFFECTS

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Abstract—Macroscopic natural convective effects in directional dendritic solidification of Al–Cu alloys are studied experimentally as a function of density gradient orientation in the interdendritic and bulk liquid relative to the gravity vector. Diffusion controlled experiments in a microgravity environment are used as a reference. Three different types of macrosegregation correspond to three different convective modes. A scaling analysis is performed in order to give a general criterion of the transition from diffusive to convective transport conditions. Our work confirms that a large number of the experimental studies previously performed to analyse primary dendrite spacings or structural transitions, have been affected by convection.

Résumé—Une étude expérimentale systématique des effets macroscopiques de la convection naturelle sur la solidification dendritique dirigée d'alliages Al–Cu a été effectuée en fonction de l'orientation, par rapport à la gravité, des gradients de densité dans le liquide interdendritique et le liquide volumique. Trois types de macroségrégations correspondant à trois modes convectifs différents ont été observés. Une analyse en ordre de grandeur des phénomènes a été effectuée dans le but d'établir un critère général prévoyant la transition d'un régime diffusif à un régime convectif. Notre travail montre qu'un nombre important de travaux expérimentaux relatifs à l'analyse des transitions structurales ou à la mesure des espacements dendriques primaires, ont eté affectés par la convection.

Zusammenfassung—Die makroskopischen natürlichen konvektiven Einflüsse beim gerichteten dendritischen Erstarren von Al–Cu-Legierungen werden experimentell in Abhängigkeit von der Orientierung des Dichtegradienten in der Flüssigkeit, und zwar außerhalb und zwischen den Dendriten, bezüglich des Schwerkraftvektors untersucht. Zum Vergleich werden diffusionsgesteuerte Experimente unter Mikrogravitation herangezogen. Drei verschiedene Arten der Makrosegregation entsprechen drei unterschiedlichen Konvektionsmoden. Um ein allgemeines Kriterium für den Übergang vom diffusiven zum konvektiven Transport zu erhalten, wird eine maßstäbliche Auswertung durchgeführt. Diese Arbeit bestätigt, daß eine große Anzahl der früher durchgeführten experimentellen Untersuchungen zum Abstand primärer Dendriten oder struktureller Übergänge durch Konvektion beeinflußt worden sind.

1. INTRODUCTION

Recent progress has been made in the understanding of dendritic solidification under purely diffusive transport conditions in the liquid phase. However, solutal convection is well known to be the source of various segregation phenomena (macrosegregation, freckles) observed in castings [1–4]. Macrosegregation has also been observed in a number of directional solidification experiments [5, 6]. The macroscopic deformation of the dendritic front observed in hypo-eutectic Al–Cu alloys solidified upwards [7–8], was attributed to solutal convection. A similar effect was recently independently reported for the case of Pb–Pd alloys solidified near the dendritic/eutectic transition [9].

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However, up till now, no consistent description of the various convective effects mentioned separately could be given.

Theoretical treatments of these convective effects have mainly consisted in numerical analysis models of macrosegregation along directions parallel to the solidification front [2, 4, 10]. A quantitative fit could not be achieved with experiments owing to the lack of reliable permeability data on real solidification conditions [3]. In addition, such effects as the appearance of freckles, macroscopic deformation of the dendritic front or longitudinal segregation, cannot be accounted for by these models since they only assume a coupling of the liquid fraction in the mushy zone with the flow. For these reasons, it was recently recognized that a more analytical approach would be useful to get a better insight of this type of phenomena [11, 12].

In this respect, the study presented here consists of the two following complementary aspects [30]:

- Systematic directional solidification experiments were performed in relation to orientation relative to the gravity vector $\boldsymbol{g}$, under otherwise identical solidification conditions. Table 1 shows the six different convective configurations as a function of density gradient orientations in the bulk and in the interdendritic liquid. A reference experiment was also performed in a microgravity environment under the same thermal conditions. In each case, both radial and longitudinal macrosegregations were determined.
- A scaling analysis was carried out to determine the characteristic dimensionless quantities controlling the natural interdendritic flow and its effects on segregation.

Table 1. Convective configurations in directional dendritic solidification: ⇧ thermal gradient (stabilizing case); ↑ solutal gradient (stabilizing case)

Relative density of the rejected species	Lower	Higher
Solidification direction: horizontal	⇨	⇨
	⇦	⇨
downwards	⇩↑	⇩↓
upwards	⇧↓	⇧↑

2. EXPERIMENTAL

2.1. Experimental method

The various configurations in Table 1 were obtained by choosing hypo- and hyper-eutectic compositions of the same binary system Al–Cu, which has a roughly symmetrical phase diagram around the eutectic Al–Al_2Cu. Alloys of compositions 26 wt% (or 20 wt%) and 40 wt% were prepared by conventional casting techniques and their homogeneity verified by density measurements.

The microgravity experiment was performed in the G.H.F. furnace (Gradient Heating Facility devised by CNES) during the D-1 Spacelab mission (Oct. 31–Nov. 6, 1985). Most of the ground-based experiments was performed in a thermal mock-up of this furnace which operated under vacuum and consisted mainly of three independently programmed heating zones. Some complementary experiments were also carried out in similar thermal conditions in a classical horizontal Bridgman furnace where efficient quenching of the solidification interface could be obtained.

The samples are inserted into a cartridge. The internal arrangement of this cartridge is schematically shown on Fig. 1 together with the initial temperature profile in the sample. The crucible consists of a boron nitride tube (I.D. = 9.5 mm, O.D. = 12.5 mm, length 130 mm).

Preliminary experiments were performed with various thermal histories in order to determine the conditions (in terms of initial melting and stabilization as well as solidification parameters) under which convective effects could best be analysed. The temperature cycle applied in the space and reference ground-based experiments is shown in Fig. 2. The

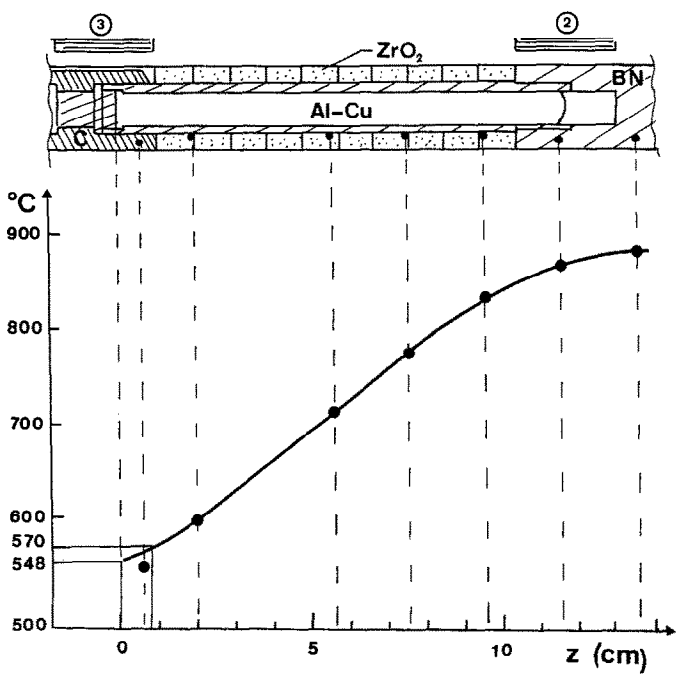


Fig. 1. Experimental set-up and initial temperature profile in the sample (② and ③: heating zones, BN: boron nitride, C: graphite).

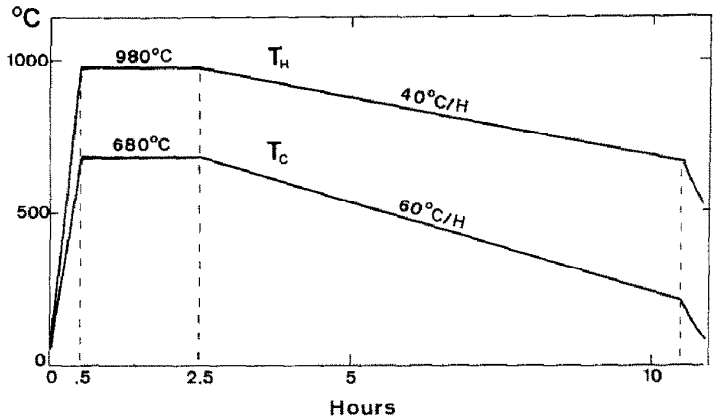


Fig. 2. Temperature cycle applied in the solidification experiments.

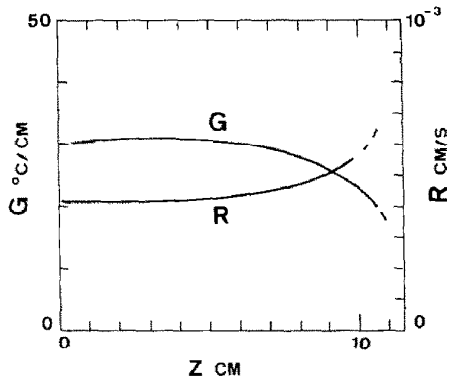


Fig. 3. Solidification rate R and thermal gradient G in the liquid in front of the interface as a function of the solidified length (case of space and reference ground-based experiments).

resulting solidification rate R and thermal gradient G in the liquid in front of the interface are given in Fig. 3 as a function of solidified length. In the part of the sample located between the heating zones 2 and 3 (Fig. 1), stationary solidification conditions were achieved with $G = 30\,\mathrm{K\,cm^{-1}}$ and $R = 4.2 \times 10^{-4}\,\mathrm{cm\,s^{-1}}$. Interface demarcations and numerical simulations indicated that, in the same region, the radial temperature difference between the centre and the wall at the solid–liquid interface never exceeded 2 K.

The solidification conditions used in the present study are summarized in Table 2.

After radiographic examination, the solidified samples were cut into 5–10 mm thick slices. Macrosegregation along the solidification direction (longitudinal segregation) was determined from density measurements of these slices using the following relation between concentration C and density ρ [13]

$$C\,(\mathrm{wt\%\,Cu}) = \frac{-379.64}{\rho\,(\mathrm{g\,cm^{-3}})} + 140.27. \qquad (1)$$

The resulting accuracy on C is better that 1 wt%. Cross-sections were then examined by classical metallography and quantimetry in order to determine the macrosegregation parallel to the solidification front (radial segregation) and investigate the macroscopic defects of the dendritic array.

2.2. Results—initial segregation effects

Longitudinal segregations were first examined on samples solidified at higher solidification rates by passive cooling of the furnace ($300\,\mathrm{K\,h^{-1}}$). Typical results are given on Fig. 4(a) and (b) where the scaled concentration $c = (C - C_E)/(C_0 - C_E)$ (to be used from now on, with C_0 and C_E respectively nominal and eutectic composition) is reported as a function of the solidified fraction f_S. These experiments show that residual longitudinal segregation (corresponding to a stable density stratification) appears in the following situations:

(a) In the solutally destabilizing cases (△, ▼) and (▽, ▼), primary phase enrichment occurs at the cold end of the sample [Fig. 4(a)].

(b) In the solutally and thermally stabilizing case (△, ▲), when the whole sample is heated above the eutectic temperature T_E, a quasi-eutectic region appears in the bottom of the

Table 2. Solidification conditions used in the present study

Furnace	G ($\mathrm{K\,cm^{-1}}$)	R ($\mathrm{cm\,s^{-1}}$)
Horizontal Bridgman with quenching	40	2.5×10^{-4} 1.25×10^{-3}
Gradient heating furnace		
Experiments with passive cooling	~30	$\sim 10^{-2}$
Space and reference ground-based experiments	30	4.2×10^{-4}

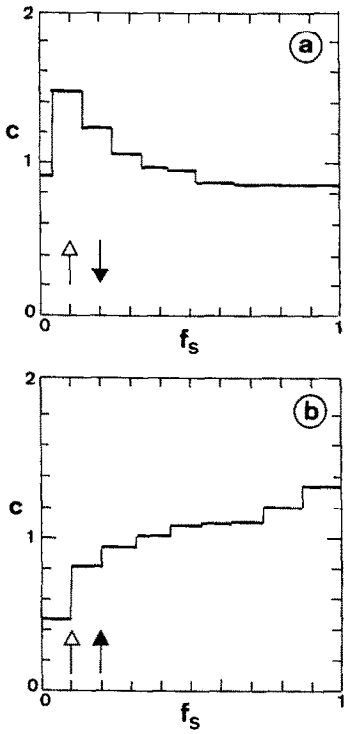


Fig. 4. Longitudinal segregation in samples solidified upwards by passive cooling of the furnace ($R \sim 10^{-2}\,\mathrm{cm\,s^{-1}}$). (a) Hyper-eutectic alloy. (b) Hypo-eutectic alloy (20 wt% Cu).

sample while the upper part is Al enriched [Fig. 4(b)].

Clearly, these segregations are due to gravity since they do not occur in space samples (Fig. 10). But they are not associated with convective effects during solidification since they are not reduced when the solidification rate is increased (see below, for comparison, the segregations in samples solidified at low rate). These segregations thus result from the effect of gravity during melting, i.e. from the settling (or floating) of dendrites during this stage.

In case (a), dendrites tend to accumulate at the eutectic front during melting and rehomogenization through the dendritic region is very slow. Since the furnace used in the present study does not allow initial mixing or overheating, this initial segregation effect cannot be avoided in our experiments and has to be taken into account in the data analysis.

In case (b), dendrites tend to move towards the bulk liquid. The effect is most drastic when the eutectic isotherm reaches the bottom of the crucible, since the entire mushy zone is then suddenly detached from the bottom and moves up to the top end of the sample, which gives the characteristic concentration profile of Fig. 4(b). Thus, in our reference experiments, the cold end of the samples was maintained below T_E, which reduces this effect to a negligible level (see Fig. 6). Although similar dendrite movement is expected in case (↓, ↑), this does not lead to significant segregation even when the entire sample is heated above T_E: homogenization is then promoted by the strong bulk convection prevailing in this thermally destabilizing configuration.

2.3. *Results—segregations induced by natural convection*

When the solidification rate is decreased, increasingly greater segregation effects can be observed, associated with convective transport during solidification. Figure 5(a) and (b) give the longitudinal segregation measured on the hyper-eutectic alloy solidified horizontally at two different rates, and quenched. Let C_S be the composition of the solid formed at a given instant from a liquid at composition C_∞. An effective partition ratio can be defined by

$$k_{\mathrm{eff}} = (C_S - C_E)/(C_\infty - C_E).$$

This quantity can be directly measured at the quenched interface. However, considering as a first approximation that k_{eff} remains constant during

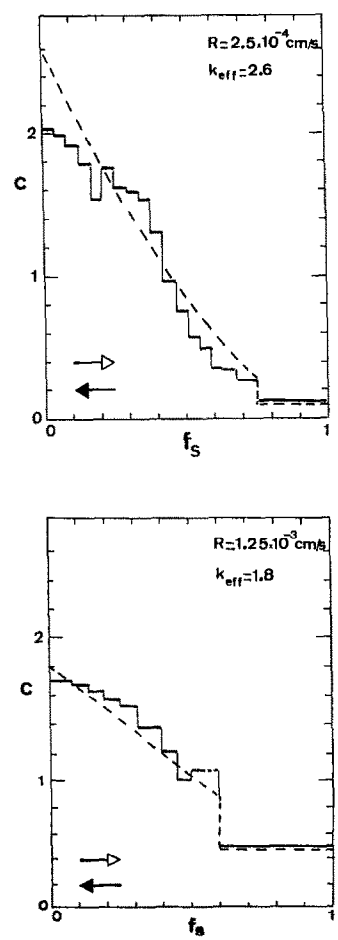


Fig. 5. Longitudinal segregation in the hyper-eutectic alloy solidified horizontally at two different rates: (a) $R = 2.5 \times 10^{-4}\,\mathrm{cm\,s^{-1}}$, (b) $R = 1.25 \times 10^{-3}\,\mathrm{cm\,s^{-1}}$. Full segments: experimental measurements. Dasked lines: theoretical curves corresponding to the given value of k_{eff}.

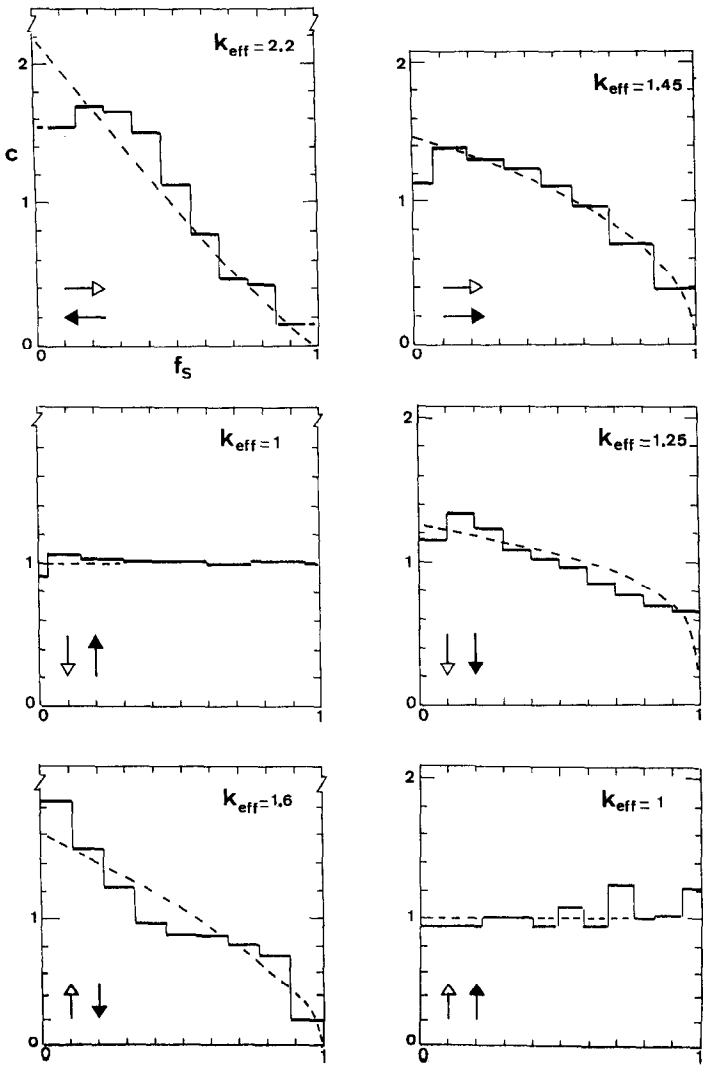


Fig. 6. Longitudinal segregation in the samples solidified in the reference conditions in the six different convective configurations of Table 1.

solidification, another estimate of this quantity can be obtained by fitting the segregation curves with a Scheil equation. Figure 5(a) and (b) show that both types of data agree reasonably well and that k_{eff} increases as R decreases.

Figure 6 shows the longitudinal segregation measured on the samples solidified in reference conditions in the six different convective configurations of Table 1. In each case, the data are fitted with a Scheil equation and the corresponding value of k_{eff} is indicated. The positive deviations from the theoretical curves observed near the cold end in cases (↑, ↓) and (↓, ↓) can be ascribed to the initial segregation effect analysed in the previous section. Typical metallographic cross-sections of the same samples are given on Fig. 7. The scaled difference $(c - c_0)$ extreme values measured by quantimetry on these cross-sections are reported on Table 3 together with the values of $k_{eff} - 1$.

Figures 6 and 7, and Table 1 show that only three types of segregation can be distinguished from the six configurations of Table 1, depending on the density gradient orientation in the interdendritic liquid relative to $\boldsymbol{g}$:

Table 3. Measured segregations in the six different convective configurations of Table 1

Alloy	Hyper-eutectic			Hypo-eutectic		
Solidification direction	$k_{eff} - 1$	$\min(c - c_0)$	$\max(c - c_0)$	$k_{eff} - 1$	$\min(c - c_0)$	$\max(c - c_0)$
Horizontal	1.2	−1	+0.7	0.45	−1	+0.3
Downwards	0	−1	+0.9	0.25	0	0
Upwards	0.6	0	0	0	−1	+0.23

Values of $k_{eff} - 1$, and extreme values of the radial concentration deviation $c - c_0$.

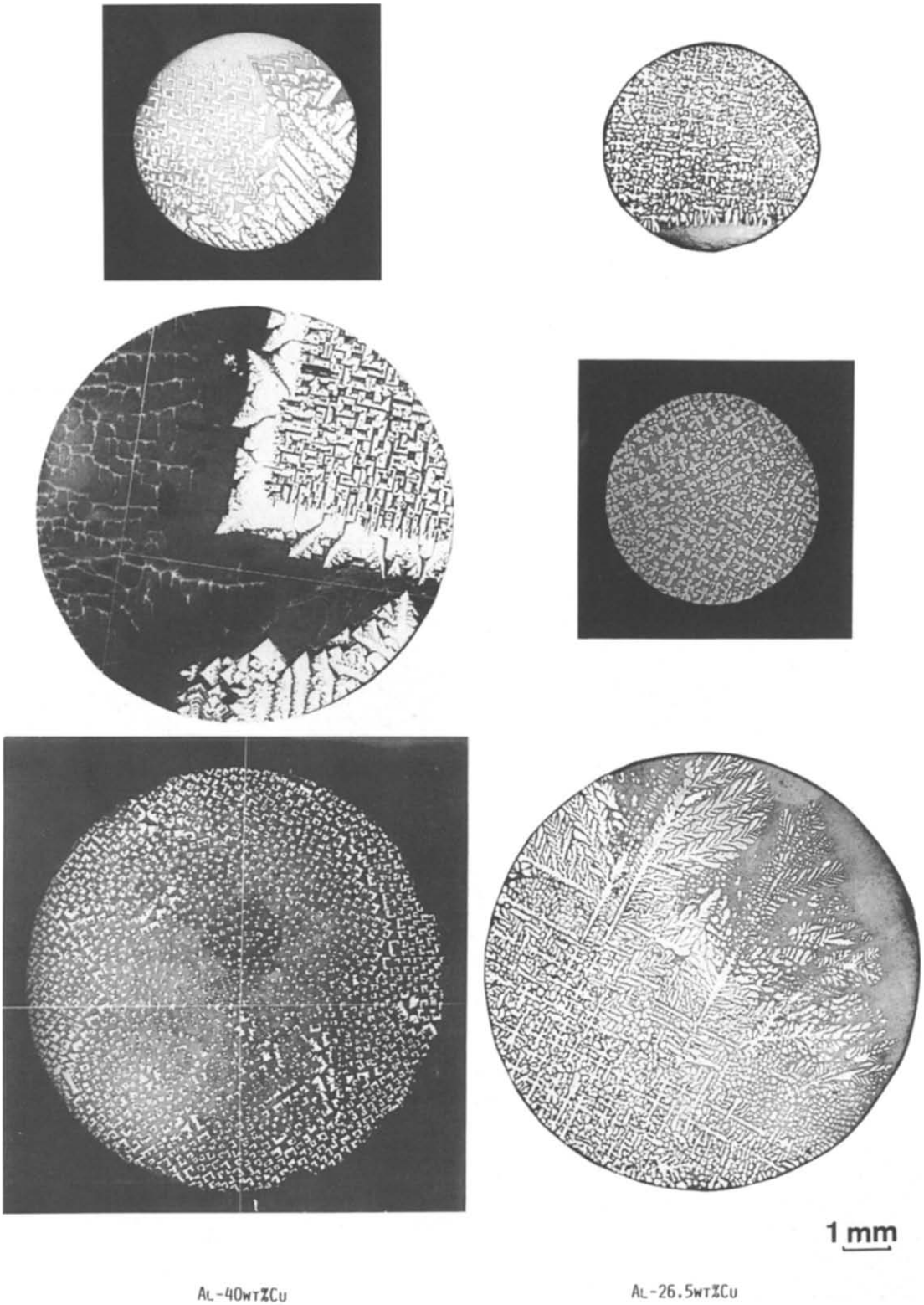


Fig. 7. Typical cross-sections (same samples as Fig. 6) showing considerable radial segregation in the cases of horizontal solidification, and vertical solidification with a solutal-stabilizing effect.

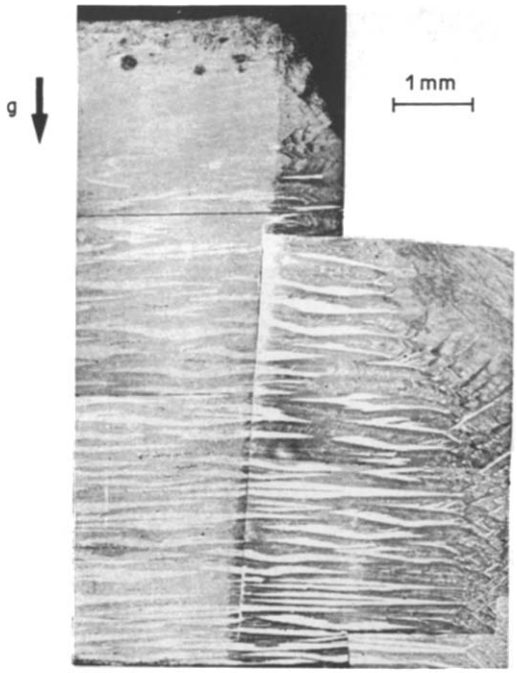


Fig. 8. Shape of the dendritic front (horizontal solidification of the hyper-eutectic alloy $R = 1.25 \times 10^{-3}\,\mathrm{cm\,s^{-1}}$).

(*a*) *Horizontal solidification.* Samples solidified horizontally show simultaneous longitudinal and radial segregation. The longitudinal segregation is more pronounced than in other configurations, so that the last solidified part is pure eutectic. The radial segregation is parallel to $\boldsymbol{g}$ and corresponds to a stable stratification. The dendritic fraction decreases (respectively, increases) as height decreases in the hypo- (respectively, hyper-) eutectic alloy and a pure eutectic layer appears at the bottom (respectively, at the top). The corresponding shape of the dendritic front appears on the vertical section taken on the growth axis of a quenched sample (Fig. 8). Dendritic growth direction is not affected in the vicinity of the eutectic layer. Finally, channel segregates are found as pure eutectic pockets on cross-sections (Fig. 9, hyper-eutectic alloy). On successive sections along the solidification direction, these pockets move upwards at an increasing rate.

(*b*) *Vertical solidification with a solutal destabilizing effect.* This mode is characterized by considerable longitudinal macrosegregation without any measurable radial macrosegregation. Channel segregates are observed on some cross-sections, although smaller than in the horizontal solidification. They generally appear at the periphery of the samples and develop along the vertical.

(*c*) *Vertical solidification with a solutal stabilizing effect.* This mode is characterized by considerable radial macrosegregation without any measurable longitudinal macrosegregation. A large dendrite-free region is found on cross-sections, a feature repeatably observed. The location of this region is not correlated with the symmetry of the thermal conditions applied to the sample. Moreover strong tendency for dendrites to grow towards the eutectic region was observed.

2.4. Results—reference microgravity experiments

Figure 10 gives the longitudinal segregation and the micrograph of a typical cross-section for the reference samples solidified under microgravity conditions. In addition to density measurements and metallographic examination, extensive quantitative image analysis was performed on these samples [14]. These data will be discussed in detail in a forthcoming paper [15]. Here, it should be noted that the longitudinal and radial macrosegregation was eliminated in the part of the samples solidified under steady reference conditions. These results agree with those of a previous microgravity experiment performed on less concentrated alloys [16].

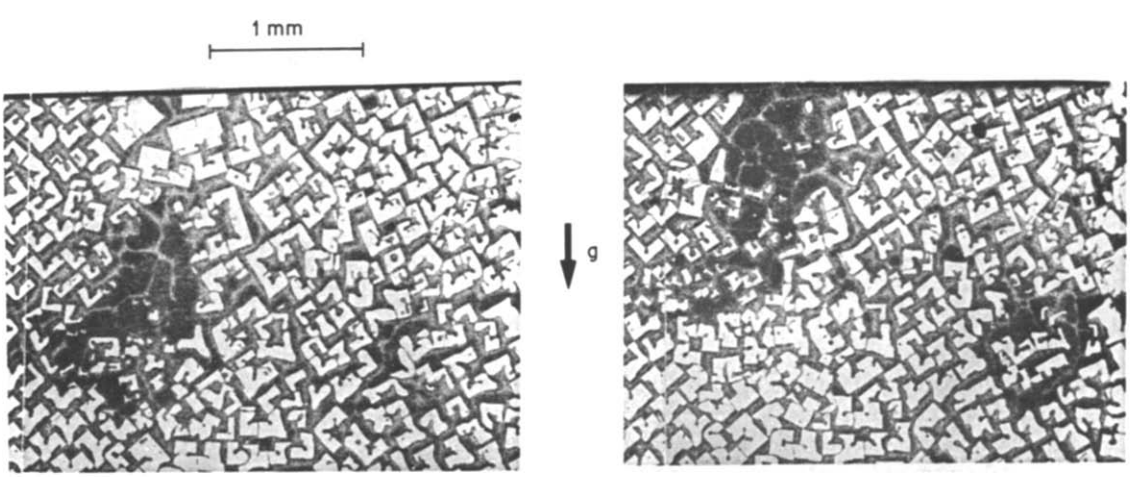
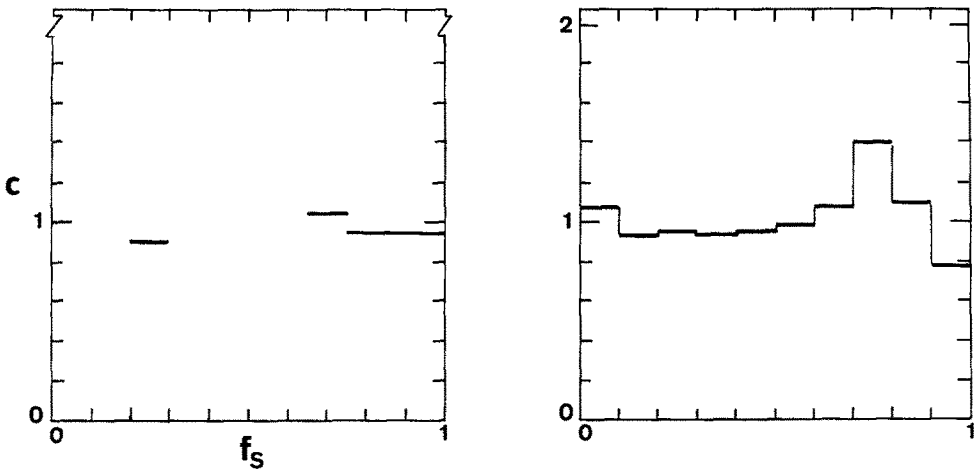
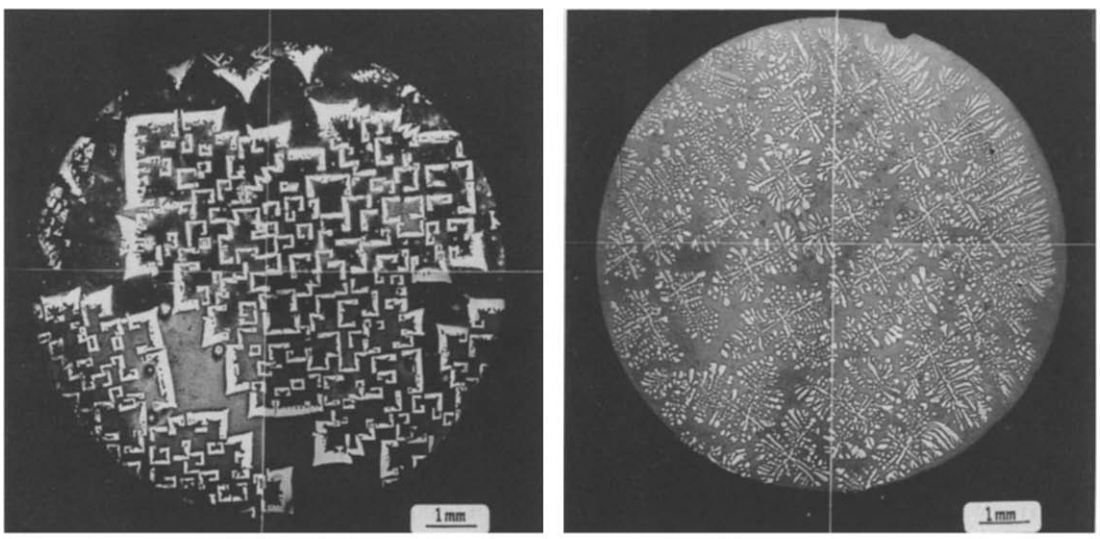


Fig. 9. Channel segregates observed on cross-sections of the hyper-eutectic solidified horizontally at $R = 1.25 \times 10^{-3}\,\mathrm{cm\,s^{-1}}$.





AL-40WT%CU μG AL-26.5WT%CU μG

Fig. 10. Longitudinal segregation and cross-sections of the reference samples solidified under microgravity conditions.

3. SCALING ANALYSIS

3.1. Physical description and governing equations

A macroscopic model based on the Flemings' hypothesis [17, 18], is considered:

- complete mixing of the interdendritic liquid up to dendrite tips,
- local thermodynamic equilibrium in this zone with no capillary or kinetic contribution;
- and continuity of the liquid concentration and concentration gradient at dendrite tips.

This description is extended here up to the limits of solidification with a planar front as well as coupled eutectic growth, which is equivalent to considering a constitutional supercooling criterion to describe stability thresholds in both cases. In addition, possible effects of convection in the bulk liquid (particularly, shifting of stability thresholds [19]) are simply accounted for through a solute boundary layer model.

Finally, only steady unidirectional solidification conditions will be considered, i.e. uniform thermal gradient G and solidification rate R, and imposed concentration C_∞ in the bulk liquid far from the interface. Since the solidification shrinkage has no effect on segregation in such conditions, density variations will be taken into account only in the momentum equations.

With these hypotheses, the system can be separated into three zones (bulk liquid, mushy zone and solid) separated by mathematical boundaries, as shown on Fig. 11, where $\mathbf{n}$ is the unit vector parallel to the temperature gradient. The resulting equations and associated boundary conditions are given below.

In the mushy zone, the local unknown quantities are: the mean interdendritic flow velocity $\mathbf{V}$, the mean

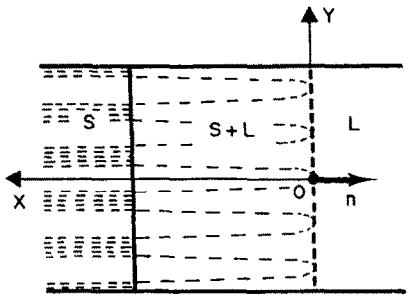


Fig. 11. Macroscopic model of the solidification front.

concentration C, the average composition of the solid phase C_α, and the liquid fraction f_L (or solid fraction f_S). The last three quantities are linked by

$$C = f_L C_L + f_S C_\alpha \tag{2}$$

and

$$\mathbf{n}\cdot\nabla C = f_L \frac{G}{m_L} + (1-k)\, C_L \mathbf{n}\cdot\nabla f_L \tag{3}$$

where C_L is the concentration of the liquid phase, m_L is the liquidus slope, k is the equilibrium partition ratio, equation (3) is derived from (2) by differentiating and using the local thermodynamic equilibrium between solid and liquid, i.e.

$$\nabla C_L = \frac{G}{m_L}\mathbf{n} \tag{4}$$

$$\mathbf{n}\cdot\nabla(f_S C_\alpha) = k C_L \mathbf{n}\cdot\nabla f_S. \tag{5}$$

In a reference frame linked with the solid phase, the balance equations for mass, solute and momentum in the mushy zone are respectively written

$$\nabla\cdot\mathbf{V} = 0 \tag{6}$$

$$\nabla\cdot[D f_L \nabla C_L) - C_L \mathbf{V}] = \frac{\partial C}{\partial t} \tag{7}$$

$$\frac{\rho_L \nu}{K}\cdot\mathbf{V} + \nabla P - \rho_L \mathbf{g} = 0 \tag{8}$$

where t is time, D the chemical diffusion coefficient in the liquid phase, ν the kinematic viscosity, K the local permeability of the mushy zone, P the total pressure, and ρ_L the local density of the liquid.

From the steady conditions hypothesis made

$$\frac{\partial C}{\partial t} = -R\mathbf{n}\cdot\nabla C. \tag{9}$$

Thus, using (4) and (9), equation (7) transforms to

$$\frac{DG}{m_L}\mathbf{n}\cdot\nabla f_L + \frac{G}{m_L}\mathbf{n}\cdot\mathbf{V} - R\mathbf{n}\cdot\nabla C = 0. \tag{10}$$

Introducing the dynamic pressure $P' = P - \rho_L \mathbf{g}\cdot\mathbf{X}$, and the solutal expansion coefficient β_c, and using a Boussinesq approximation, equation (8) becomes

$$\frac{\rho_L \nu}{K}\mathbf{V} + \nabla P' - \rho_{L_0}\beta_c(C_L - C_{L_0})\mathbf{g} = 0. \tag{11}$$

Taking into account the equivalence between a porous medium and a classical Hele–Shaw cell, it will be assumed that

$$K = B(f_L \lambda)^2 \tag{12}$$

where B is a dimensionless constant, and λ is a characteristic microscopic dimension of the mushy zone, i.e. the primary interdendritic spacing.

Along the eutectic isotherm, the remaining liquid solidifies as a eutectic at concentration C_{SE}. The corresponding boundary conditions are written

$$\begin{aligned} \mathbf{V}\cdot\mathbf{n} &= 0 \\ C_{\alpha-} - C_{\alpha+} &= 0 \\ D\mathbf{n}\cdot\nabla C + R(C_L - C_{SE}) &= 0 \end{aligned} \tag{13}$$

where subscripts $-$ and $+$ denote the value of a quantity on either side of the boundary.

The boundary conditions along the surface separating the mushy zone and the bulk liquid are simple continuity conditions

$$\begin{aligned} \mathbf{V}_- &= \mathbf{V}_+ \\ C_{L-} &= C_+ \\ \frac{G}{m_L}\mathbf{n} &= (\nabla C)_+ \\ P'_- &= P'_+. \end{aligned} \tag{14}$$

Note that, in the general case, these boundary conditions couple the concentration as well as pressure and velocity fields in the mushy zone and bulk liquid.

Finally, conditions in the bulk liquid and at the crucible walls can be described by classical balance equations, as found for instance in [20].

3.2. Reference solution in the absence of interdendritic flow

Consideration is given first to the solution of the model where there is no convective motion in the mushy zone, restricted to the case $k < 1$. In as much that a unidimensional description of solute transport in the bulk liquid can be adopted, the dendritic front is planar. From now on its position will be considered as the origin of the X axis antiparallel to the solidification direction. The solute concentration $C_L(0)$ at dendritic tips is given, similarly to the case of monophase growth [20], by

$$\frac{C_L(0)}{C_\infty} = \frac{1}{1-(1-k)\Delta} \tag{15}$$

with

$$\Delta = \min\left(1, \frac{R\delta}{D}\right). \tag{16}$$

Here, δ is the convective solute boundary layer whose expression depends on the type of external convection involved (thermal buoyancy or forced convection). Following [21] this concept will be ex-

tended to the case where mixing is due to the solutal buoyancy originating in the boundary layer itself.

Introducing (15) in the equations, the domains of the three different types of solidification structure (monophase, eutectic, dendritic) are easily derived. Dendritic structures occur only above the critical rate

$$R_c = \frac{DG}{|m_L|(1-k)C_E} \tag{17}$$

and the transitions between structures are defined by the following relations

- monophase/eutectic transition: $R < R_c$

and

$$C_\infty^0 = C_E[1-(1-k)\Delta] \tag{18}$$

- monophase/dendritic transition: $R > R_c$

and

$$C_\infty^* = \frac{DG[1-(1-k)\Delta]}{R|m_L|(1-k)} \tag{19}$$

- dendritic/eutectic transition: $R > R_c$

and

$$C_\infty^{**} = C_E - \frac{DG}{R|m_L|}\Delta. \tag{20}$$

In the dendritic structure domain, the liquid fraction in the mushy zone is given by

$$f_L(X) = \left[\frac{GX}{|m_L|(C_\infty - C_\infty^*)} + 1\right]^{-1/(1-k)} \tag{21}$$

with: $0 < X < X_E$

where X_E is the total length of the mushy zone:

$$X_E = \frac{|m_L|}{G}(C_\infty^{**} - C_\infty). \tag{22}$$

Choosing the characteristic length $G/|m_L|(C_\infty - C_\infty^*)$, a dimensionless form is obtained

$$f_L(x) = (x+1)^{-1/(1-k)} \tag{23}$$

with

$$0 < x < \frac{C_\infty^{**} - C_\infty}{C_\infty - C_\infty^*}. \tag{23}$$

3.3. *First order contribution of the interdendritic flow*

The first order contribution to the interdendritic flow is the solution of the problem defined by neglecting the coupling of the permeability K (i.e. of f_L and/or λ) and boundary conditions, with the flow. For the sake of simplicity, a bi-dimensional configuration is considered here with a crucible of width H inclined at an angle θ relative to the gravity vector. The problem is thus a generalization of the one considered in [11].

If X^* is a characteristic length scale of the mushy zone (to be chosen below), pressures can be scaled by $\rho_{L_0} g\beta_c[C_L(X^*) - C_L(0)]X^*$ and velocities by $g\beta_c[C_L(X^*) - C_L(0)]K(X^*)/\nu$. The corresponding dimensionless quantities will be designated: x, y, p and $\mathbf{v}$. The following dimensionless form of the problem is then obtained

$$u = \mathscr{F}(x, k, \mathscr{A}_1)\left[\frac{\partial p}{\partial x} - x\cos\theta\right]$$
$$v = \mathscr{F}(x, k, \mathscr{A}_1)\left[\frac{\partial p}{\partial y} - x\sin\theta\right] \tag{24}$$

with

$$0 < x < \mathscr{A}_2$$
$$-\mathscr{A}_3 < y < +\mathscr{A}_3$$

and the boundary conditions

$$p = 0 \quad \text{at} \quad x = 0$$
$$u = 0 \quad \text{at} \quad x = \mathscr{A}_2 \tag{24'}$$
$$v = 0 \quad \text{at} \quad y = -\mathscr{A}_3 \quad \text{and} \quad y = +\mathscr{A}_3.$$

$\mathscr{A}_1$, $\mathscr{A}_2$ and $\mathscr{A}_3$ are three length ratios defined by

$$\mathscr{A}_1 = \frac{|m_L|(C_\infty - C_\infty^*)}{GX^*},$$
$$\mathscr{A}_2 = \frac{|m_L|(C_\infty^{**} - C_\infty)}{GX^*} \quad \text{and} \quad \mathscr{A}_3 = \frac{H}{2X^*}.$$

And $\mathscr{F}(x, k, \mathscr{A}_1)$ is a permeability ratio which, using (12) and (23), is found to depend on k and $\mathscr{A}_1$

$$\mathscr{F}(x, k, \mathscr{A}_1) = \left(\frac{1+\mathscr{A}_1}{x+\mathscr{A}_1}\right)^{2/(1-k)}. \tag{25}$$

Since one of the three length ratios ($\mathscr{A}_1$, $\mathscr{A}_2$ and $\mathscr{A}_3$) can be fixed by a proper choice of X^*, the flow field finally depends on three parameters: two length ratios, and the equilibrium partition ratio.

The limiting case $H \to \infty$ is considered first. Here, as found in [11], the flow is unidirectional and parallel to the isotherms. The flow field is found to depend explicitly on the experimental parameters

$$v(x, k, \mathscr{A}_1) = x\mathscr{F}(x, k, \mathscr{A}_1)\sin\theta$$
$$0 < x < \mathscr{A}_2. \tag{26}$$

Choosing the length scale $X^* = |m_L|(C_\infty - C_\infty^*)/G$ (i.e. $\mathscr{A}_1 = 1$), the velocity profile depends only on k.

Asymptotically similar solutions are obtained for $k \to 1$ by choosing $X^* = |m_L|(C_\infty - C_\infty^*)(1-k)/G$ [i.e. $\mathscr{A}_1 = 1/(1-k)$], since then

$$F(x, k) \to e^{2(1-x)} \quad \text{for} \quad k \to 1.$$

But a convenient practical length scale is

$$X_{max} = \frac{|m_L|(C_\infty - C_\infty^*)}{G}\frac{(1-k)}{(1+k)}$$
$$\left(\text{i.e. } \mathscr{A}_1 = \frac{1+k}{1-k}\right) \tag{27}$$

which ensures that $v(x)$ goes through the maximum value 1 at $x = 1$ for any k value. Figure 12 gives the

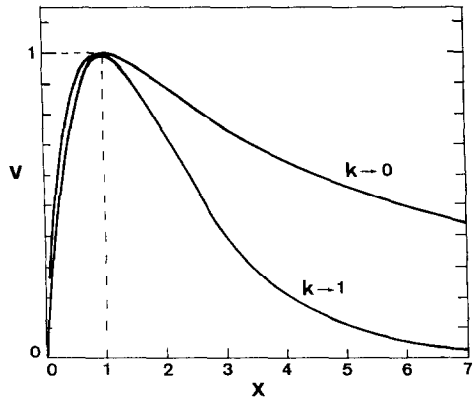


Fig. 12. k dependence of the velocity profile in the flow configuration 1 of figure 13.

corresponding velocity profiles, with the two limiting cases

$$v(x) \to x\left(\frac{2}{1+x}\right)^2 \quad \text{for} \quad k \to 0$$

$$v(x) \to xe^{1-x} \quad \text{for} \quad k \to 1. \tag{28}$$

This figure shows that the flow will be confined in the region where x is of order one. This is especially true since the permeability should be lower than predicted by equation (12) in the regions far from dendrite tips (liquid fractions being generally lower than theoretical values, and coarsening effects leading to isolated liquid pools).

The same type of analysis can be applied to discuss the possible effect of a very rapid variation in permeability at high liquid fractions, as has been suggested in the literature [23].

For example, assuming

$$K(X) = B_n f_L^n(X)\lambda^2 \tag{29}$$

then, correcting the length scale to

$$X^* = \frac{|m_L|(C_\infty - C_\infty^*)}{G}\,\frac{(1-k)}{n-(1-k)} \tag{30}$$

the following relation is obtained

$$v(x) \to xe^{1-x} \quad \text{for} \quad \frac{1-k}{n} \to 0 \tag{31}$$

which is identical to the curve of Fig. 12 corresponding to $k \to 1$.

Thus, a high value of n would lead to maximum flow velocity very near to dendrite tips where the liquid fraction is near to one. This might be of little significance, since in this region the hypothesis of complete mixing is poor, and thus the local driving force for convection is less than anticipated.

For these reasons, the length scale finally retained is X_{max} given by (27). Two limiting cases should be distinguished depending on the relative value of X_{max} and the length of the mushy zone X_E:

(1) If $X_{max} \ll X_E$, the flow is confined to a region of extent X_{max} and does not depend on X_E (case 1 of Fig. 13).

(2) If $X_E \ll X_{max}$, the flow increases linearly from zero at dendrite tips to its maximum value at the eutectic isotherm (case 2 of Fig. 13). The proper length scale X^* is then X_E (i.e. $\mathscr{A}_2 = 1$).

The last asymptotic situation corresponds to the case where the crucible width H is much smaller than both X_{max} and X_E (case 3 of Fig. 13). The flow is then parallel to the crucible walls, and scaling the length by $H/2$ (i.e. $\mathscr{A}_3 = 1$) gives

$$u = y\mathscr{F}\left(x, k, \frac{2m_L(C_\infty - C_\infty^*)}{GH}\right)$$

$$-1 < y < 1 \tag{32}$$

with

$$\mathscr{F} \simeq 1 \quad \text{when} \quad x \ll \frac{2|m_L|(C_\infty - C_\infty^*)}{GH}.$$

Finally, the above analysis gives the following scaling for the first order contribution to the interdendritic flow velocity

$$V^* = B\frac{\beta_c g}{\nu}\cdot\frac{G}{|m_L|}X^* \times\left[\frac{GX^*}{m_L(C_\infty - C_\infty^*)} + 1\right]^{-2/(1-k)}\cdot\lambda^2\sin\theta$$

with

$$X^* = \min\left[\frac{|m_L|(C_\infty - C_\infty^*)}{G} \times\frac{(1-k)}{(1+k)}, \frac{H}{2}, \frac{|m_L|}{G}(C_\infty^{**} - C_\infty)\right]. \tag{33}$$

In addition, the velocity fields corresponding to the three asymptotic domains are the unidirectional ones shown on Fig. 13. This first order contribution to flow is cancelled when the temperature gradient is parallel to the gravity vector.

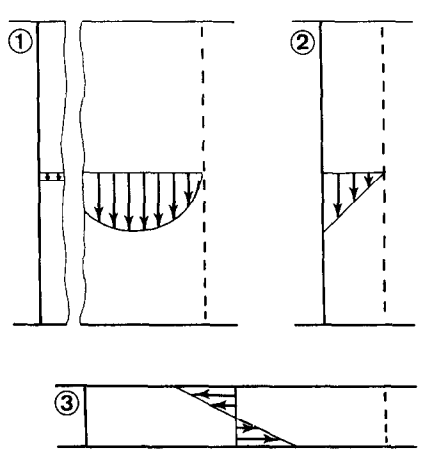


Fig. 13. Asymptotic configurations for the interdendritic flow during horizontal solidification: 1. infinitely thick sample, low concentration; 2. infinitely thick sample, high concentration; 3. thin sample.

In real situations, where the crucible can be large but not infinite, the flow should turn (respectively inside and outside the mushy zone) in end regions, the extent of which should be of the order of X^*.

The concentration field in the mushy zone C can be written as the sum of $C_0(X)$ obtained in diffusive transport conditions and $\tilde{C}(X, Y)$ resulting from the contribution of interdendritic flow. Replacing C in this manner in (10) gives

$$\mathbf{n}\cdot\nabla\tilde{C} = \frac{G}{|m_L|R}\mathbf{n}\cdot\mathbf{V}. \tag{34}$$

Assuming a non-dimensional concentration defined by

$$\tilde{c} = \frac{|m_L|\tilde{C}}{GX^*} \tag{35}$$

then, according to (34), $\tilde{c}$ depends on the parameter

$$\Gamma = \frac{V^*}{R} \tag{36}$$

and the geometry of the configuration.

If the flow is parallel to the isotherms ($\mathbf{n}\cdot\mathbf{V} = 0$) (configurations 1 and 2 of Fig. 13), no segregation is induced. In these cases, segregation will occur only in the turning end regions where

$$\tilde{c} \sim \Gamma. \tag{37}$$

For configuration 3 of Fig. 13, the segregation is given by

$$\tilde{c} = xy\Gamma \quad \text{for} \quad x \ll \frac{2|m_L|(C_\infty - C_\infty^*)}{GH}. \tag{38}$$

Thus, for all cases, the dimensionless parameter Γ defined by (33) and (36) gives a measure of the first order effect of interdendritic flow on segregation. In the vertical configuration, only second order effects can occur which would be associated with a deformation of the dendritic front, i.e. coupling with the transport processes in the bulk liquid.

4. DISCUSSION

4.1. Convective modes

The experimental results obtained, a consistent interpretation can be made of the different types of macroscopic convective effects which have been separately mentioned in the literature. These are found to be controlled by orientation of the solutal density gradient in the interdendritic liquid relative to the gravity vector, independently of the density gradient in the bulk liquid. This shows that convection is mainly of solutal origin. Moreover, it is striking that the segregation effects of the various configurations are of comparable amplitude although of different type. This suggests that the average flow intensities are similar, but the flow structures are different. The three different types of flows corresponding to the three different types of segregation are described below.

1. In the horizontal configuration, there is simultaneous local solute transport and mixing with the bulk. A unique cell is formed, the direction of movement being, for instance, upwards in the case of the hyper-eutectic alloy, since the interdendritic liquid is then lighter. This flow enters the bulk liquid where it is mixed. Thus, the interdendritic flow velocity V_i and average exchange velocity are of the same order. The resulting local Al enrichment causes the observed deformation of the dendritic front. Part of the interdendritic flow concentrates into liquid pockets which in turn move upwards at an increasing rate as they get larger. According to [2], these pockets are not generated at the bottom of the sample but rather in the upper region where the flow begins to go away from the interface.

2. In the vertical solutal-destabilizing configuration, the considerable longitudinal segregation shows that there is an average exchange of the same order as in the horizontal case. However, since there is no radial macrosegregation, the motion is not macroscopically organized, but occurs through the intermittent formation of segregated channels. These channels cannot be generated within the dendritic zone as there is no driving force for convection in this zone except the residual horizontal temperature differences. These are very low in experiments considered here, and so could only cause a much slower motion than in the horizontal case. Thus, in accordance for instance with [22], segregated channels should be nucleated at the dendritic front as a result of local instability which may couple with local deformation of the front. The average exchange finally obtained is comparable to that observed in the horizontal case since the development of these channels is controlled by their feeding through the mushy zone.

3. Finally, in the vertical solutal-stabilizing configuration, the flow is as described in [7] and [8] for the particular case of upward solidification: the interdendritic liquid moves towards any depressed region of the solidification front, which further retards solidification there. In addition, it is found that the macroscopic convective cell finally formed does not mix with the bulk liquid since no longitudinal macrosegregation occurs.

4.2. Criterion for the diffuso-convective transition

As already noted segregations of comparable amplitudes were observed in the experiments for the three different convective configurations described above. Thus, from a practical point of view, the transition from diffusive to convective transport conditions can be considered to occur in the same range for all three configurations. According to the theoretical analysis given above, this transition is defined in the case of horizontal solidification, by

$$\tilde{c} \sim 1 \quad \text{i.e.} \quad \Gamma\left(\theta = \frac{\pi}{2}\right) \sim 1. \tag{39}$$

Table 4. Values of the physical data used in the evaluation of our parameter Γ'

System	Al (Cu)	Al_2Cu (Al)	Pb (Sn)
m_L (K^{-1})	340	300	233
C_E (wt/wt)	0.332		0.619
k	0.17	0	0.3
β_c (K^{-1})	1	1	0.4
ν ($cm^2 s^{-1}$)	5×10^{-3}	5×10^{-3}	2×10^{-3}
D ($10^{-5} cm^2 s^{-1}$):			
dilute alloy	5.4[27]S 5[29]D		3[29]D
off-eutectic	3.26[28]S		0.6[28]S
average value used in calculations	4	4	1

S: measured in solidification experiments.
D: measured by classical methods.

The practical evaluation of this criterion will be discussed below, and its adequacy will be checked against the results obtained and the other available experimental data.

Of the physical data necessary to evaluate the convective parameter Γ, permeability is the most difficult to obtain with good accuracy, because the dendritic structures strongly depend on the solidification conditions, and few measurements have been performed on real columnar structures. A recent compilation [24] shows that equation (12) is satisfied for flows parallel to the primary dendrite arms, with $B = 3.75 \times 10^{-4}$. This compares very well with the value 3×10^{-4} that was directly predicted from segregation data [30]. The permeability value used in the definition of Γ is in principle that of the structure assumed to be unmodified by convection, i.e. λ is the primary spacing λ_{th} obtained in pure diffusive transport conditions. Since this quantity may differ from the measured primary spacing λ_{exp} and is not necessarily known, a practical parameter Γ' will be defined from λ_{exp}.

The values of the other physical data used in subsequent calculations are summarized in Table 4. These are average values appropriate for a composition range intermediate between the monophase and the eutectic composition. The composition dependence of density and kinematic viscosity can be estimated within 2% in the Sn–Pb system by supposing an ideal behaviour [25, 26]. The accuracy obtained by this method would be much worse for systems such as Pb–Au and Pb–Pd where the working temperature is much lower than the melting point of Au and Pd.

Finally, the available data on the diffusion coefficient D are of two different origins: either classical methods, or direct measurements of the concentration profile at the quenched interface of dilute [27] or off-eutectic [28] alloys directionally solidified in a solutally stabilizing configuration. According to these data, the relative variation of D is as high as $\pm 100\%$ in the investigated range in the Pb(Sn) system. However, the temperature and/or concentration dependence of D will not be taken into account in the present first order calculations.

With the help of the data given above, the value of the parameter Γ' can be calculated for each of the experiments performed. To illustrate this, a diagram of concentration vs solidification rate has been drawn, showing the points corresponding to the experimental conditions together with the boundaries of the various domains defined in section 3. Figure 14 corresponds to the case of horizontal solidification at 1 g. It can be seen that these experiments are at the limit between configurations 2 and 3 of Fig. 13, or even slightly within the region of configuration 3 for the 20 wt% Cu alloy. Thus, the correlation between

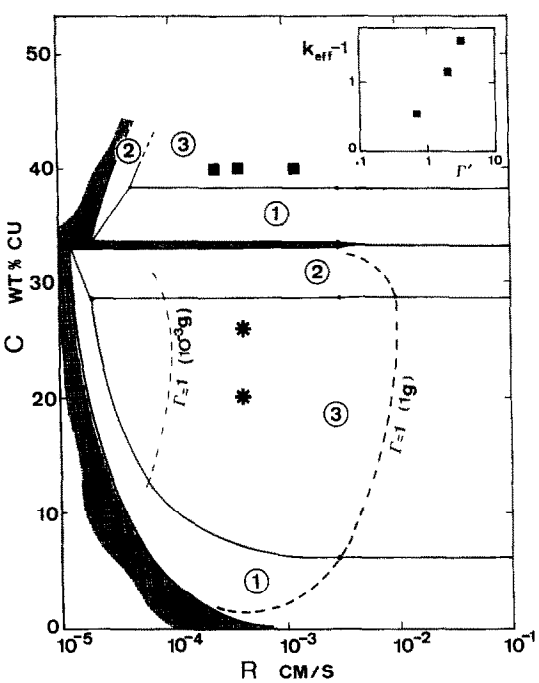


Fig. 14. Characteristic solidification domains for Al–Cu alloys with $G = 30\ K\ cm^{-1}$. Shaded areas: planar front and eutectic solidification. Unshaded area: domain of dendritic solidification with three subdomains corresponding to the three convective regimes schematically shown on Fig. 13. The dashed line separates the regions where the dendritic solidification is respectively diffusion-controlled (at higher rates) and convection-controlled (at lower rates). The squares and asterisks correspond to the experimental conditions of this study. The inserted figure shows the correlation between the segregation measured in these experiments, and the parameter Γ'.

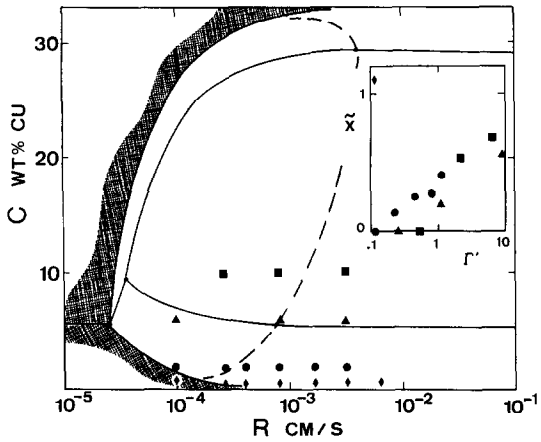


Fig. 15. Same as Fig. 14 for the experiments of Burden *et al.* [7] on interface deformation in Al (Cu) alloys solidified upwards.

the segregation data obtained and the parameter Γ' can be directly analysed in terms of the concentration scale defined in section 2 for the case of the hypereutectic alloy. It would be better to correct this concentration scale according to (35) for the 20 wt% Cu alloy. The correlation obtained can be appreciated on Fig. 14 where $k_{\text{eff}} - 1$ is plotted against the calculated Γ' values.

The correlation of the other available experimental data with the parameter Γ' is also analysed below. Figure 15 shows the points corresponding to the experiments of Burden *et al.* [7] where interface shapes have been systematically observed. Since primary spacings are not given in this paper, the values of this quantity used in the calculation of Γ' were approximately derived from [8]. Let $\tilde{X}$ be the distance between the uppermost dendrite tips and the section where the area fraction occupied by dendrites is about one-half. The planar deviation of the interface shape will be defined here as the ratio

$$\tilde{x} = \frac{\tilde{X}}{X^*}.$$

The correlation between $\tilde{x}$ and Γ is shown in Fig. 15.

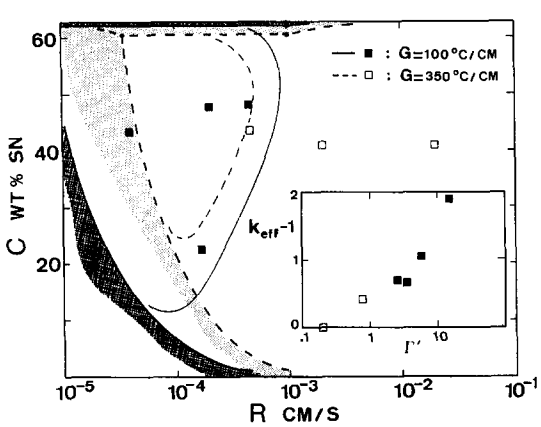


Fig. 16. Same as Fig. 14 for the experiments of Mason *et al.* [6] on primary spacings in Pb (Sn) alloys solidified upwards.

Finally, some data on segregation in a solutal-destabilizing case are given in the paper by Mason *et al.* on the measurement of primary spacings in Pb-rich Pb–Sn alloys solidified upwards. Figure 16 shows that, in this case again, a satisfactory correlation is found between the values of the quantity $k_{\text{eff}} - 1$ derived from the concentrations measured at the quenched interfaces, and the parameter Γ' used in the present study.

5. CONCLUSIONS

This experimental study has shown that the structure of the natural convective flow and its coupling with the bulk liquid essentially depend on the orientation of the density gradient in the interdendritic liquid relative to the gravity vector. However, the average flow velocity was found to be of the same order of magnitude in all cases. In particular, thermal and/or solutal-stabilizing configurations were not efficient in reducing the flow.

Scaling analysis was successfully applied to predict the transition from diffusive to convective transport conditions. Approximations were used to give a consistent description of the whole range of dendritic growth conditions between planar front instability and coupled eutectic growth. Natural flow effects were shown to be controlled by a unique characteristic dimensionless group which was explicitely defined as a function of the experimental parameters.

The study shows that natural convection probably often affected experiments on structural transitions (dendritic/eutectic and cellular/dendritic) and primary dendritic spacings. A discussion of the latter point based on the present study will be presented in a forthcoming paper.

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NOMENCLATURE

$\mathscr{A}_1, \mathscr{A}_2, \mathscr{A}_3$ = length ratios
B = dimensionless constant
C = concentration
C_L, C_0, C_∞, C_E = local, initial, bulk and eutectic liquid concentrations
C_S, C_α, C_{SE} = concentration of the two-phase solid, primary α phase, and eutectic phase
c = scaled concentration
D = solute diffusion coefficient
f_L, f_S = liquid and solid fraction
G = thermal gradient
$\mathbf{g}$ = gravity vector
H = crucible width
k, k_{eff} = equilibrium and effective partition ratio
K = local permeability of the mushy zone
m_L = liquidus slope
$\mathbf{n}$ = unit vector parallel to the temperature gradient
P, P' = total and dynamic pressure
p = dimensionless pressure
R = solidification rate
t = time
T_E = eutectic temperature
$\mathbf{V}$ = interdendritic flow velocity
u, v = dimensionless components of the interdendritic flow velocity
X, Y = coordinates
x, y = dimensionless coordinates
β_c = solutal expansion coefficient
Γ = dimensionless interdendritic convective parameter
δ = solute boundary layer extent
Δ = dimensionless bulk convective parameter
θ = angle of solidification direction
λ = primary interdendritic spacing
ν = kinematic viscosity
ρ = density